

AARADHYA DEV TAMRAKAR

Kathmandu, Nepal • aaradhyadevtmr@gmail.com • +977 9844602050 • [linkedin.com/in/aaradhya-dev-tamrakar](https://www.linkedin.com/in/aaradhya-dev-tamrakar) • github.com/AaradhyaDT • aaradhya-dev-tamrakar.github.io



PROFILE

Electronics, Communication and Information student at IOE, Kathmandu Engineering College (Year IV/I). Builds ML pipelines spanning classification, regression, tree-based ensembles, and agentic Text-to-SQL systems; deploys via FastAPI and Docker; develops Android apps with Kotlin/Jetpack Compose. Advancing through the Fusemachines AI Fellowship and NSSR DataCamp Fellowship, while leading SPARK, an ESP32-S3 edge-AI wearable fall-detection system.

EDUCATION

Bachelor of Engineering (B.E.)

Expected: January 2027

Electronics, Communication and Information — Institute of Engineering (IOE), Kathmandu Engineering College | Year IV / Part I

Coursework: Machine Learning · Signals and Systems · Digital Electronics · Control Systems · Communication Systems

EXPERIENCE

Vice Chair

2026 – Present

IEEE KEC KTM Student Branch

- Directed branch strategy, scaling initiatives and member engagement across executive committees.
- Managed relations across the broader technical community.

Event Manager

2026 – Present

Electronics Project Club (EPC), Kathmandu Engineering College

- Coordinated logistics for club workshops and department activities.

Makerspace Ambassador

June 2026 – Present

KEC Maker's Space, Kathmandu Engineering College

- Shipped site-wide fixes to the Maker's Space website — image repair, content consistency, UI cleanup.

Vice Secretary

2025 – 2026

IEEE KEC KTM Student Branch

- Streamlined operations and grew program participation by managing documentation and orchestrating workshop logistics.
- Coordinated volunteers alongside the committee for seamless execution.

Resource Manager

2024 – 2026

Electronics Project Club (EPC), Kathmandu Engineering College

- Allocated scheduling and resources for workshops across multiple departments.

FELLOWSHIPS & RECOGNITIONS

Fuse AI Fellow — Fusemachines AI Fellowship 2026

Active

Fusemachines

- Qualified through the Fusemachines AI Fellowship entrance exam (March 2026): linear algebra, calculus, probability, Python, ML.
- Developed a containerized REST API with FastAPI, PostgreSQL, and Docker.
- Engineered a five-stage agentic Text-to-SQL pipeline (Planner → Generator → Validator → Executor → Summarizer).
- Validated the five-stage Text-to-SQL pipeline at 100% execution success and accuracy on a 50-question benchmark.
- Created an end-to-end churn and CLV pipeline achieving 5-fold CV ROC-AUC 0.841, threshold-tuned for the top-200 high-risk segment.
- Trained Random Forest and XGBoost ensembles with SMOTE and SHAP explainability.
- Implemented probabilistic models spanning Bayesian estimation and Gaussian process regression via PyMC/ArviZ.
- Segmented customers on UCI Online Retail II (~500K transactions) using K-Means and DBSCAN; delivered an executive narrative.
- Benchmarked nine forecasting models; built a four-model ensemble that achieved MASE 2.44.

NSSR DataCamp Fellow — Cohort 2

2026 – Present

Nepalese Society of Student Researchers

- Secured competitive selection (April 2026) for a sponsored DataCamp Premium license, one of a limited cohort.

- Delivered six applied GenAI solutions, automating null filtering, deduplication, and schema normalization across product and data workflows.
- Completed coursework toward SQL Associate and Python Data Associate certifications in PostgreSQL, Pandas, statistics, and scikit-learn.

PROJECTS

SPARK — Wearable Fall Detection System (Major Project)

ESP32-S3 · TensorFlow Lite Micro · SHAP · MQTT · FastAPI · PostgreSQL

- Designed a two-layer ESP32-S3 fall-detection system combining a <5ms threshold gate with an INT8 1D CNN confirmation layer under 100ms.
- Computed per-event SHAP explanations at a local, non-cloud gateway.
- Presented the SPARK project proposal to the review panel; demo targeted for March 2027.
- Led the AI and backend work package (WP 1.0) — CNN training pipeline, FastAPI/PostgreSQL gateway — within a 4-person team supervised by Er. Dipen Manandhar.

Gesture-Controlled Self-Balancing Robot (GCSBR) — Minor Project

Computer Vision · Arduino · MPU6050 · Stepper Motors · Android · MATLAB

- Earned a 'major project level' rating from the examiner.
- Implemented PID stabilization firmware, validated via a MATLAB-simulation-to-hardware pipeline.
- Developed an Android gesture-control app (MediaPipe, CameraX, HC-05 BT) driving motors for real-time actuation.

Edge AI Stability Detection System

Python · Scikit-learn · FastAPI · Joblib

- Predicted platform stability from simulated IMU data using a Random Forest model, reaching 99.8% test accuracy; served via FastAPI.

Alpha Android Super-App

Kotlin · Jetpack Compose · Material3 · DataStore · Apache POI

- Assembled a modular Android super-app (SDK 36) with gesture control, an eSewa Gmail budget tracker, and a Material3 design system.

Nexus — Personal AI Operating System

React (Vite) · FastAPI · SQLite + FTS5 · httpx · asyncio

- Unified Groq (Llama 3.3 70B) and Gemini into an AI workspace via parallel fan-out routing, with SQLite + FTS5 context injection.

SKILLS

Programming: Python, C, C++, Kotlin, SQL (PostgreSQL / SQLite), VHDL

Machine Learning: scikit-learn, NumPy, Pandas, XGBoost, Random Forest, Logistic/Ridge/SGD Classifiers, Lasso, ElasticNet, SMOTE (ImbPipeline), SHAP, GridSearchCV, Stratified K-Fold CV, Joblib, PyMC, ArviZ, pgmpy, Bayesian Inference, Gaussian Process Regression, K-Means, Hierarchical Clustering, DBSCAN, Time Series Forecasting (SARIMA, Holt-Winters, Prophet, LightGBM, LSTM)

Android Development: Kotlin, Jetpack Compose, Material3, DataStore, CameraX, MediaPipe, Apache POI

Deployment: FastAPI, REST APIs, Docker, PostgreSQL, Streamlit, asyncio, Joblib

Embedded Systems: Arduino, ESP32-S3, MPU6050, Stepper Motors, HC-05 BT, UART, MQTT, FPGA, Vivado

AI / Local LLM: Ollama, AnythingLLM, Prompt Chaining, Agentic Query Systems

Tools: Git, GitHub, VS Code, Jupyter Notebook, Google Colab, MATLAB, LaTeX / Overleaf, SSMS

Other: Project Coordination, Public Relations, Microsoft Office, Canva

CERTIFICATIONS & ACTIVITIES

IEEE WIE Nepal LaTeX Workshop (May 2026) · IEEE SPAX organizing team · Prompt Engineering Fundamentals, TechAxis · IEEE Xtreme 19.0 (Team ShadowXTREME) · PCB Design & Fabrication Workshop, KEC Robotics Club · AWS Cloud Computing Workshop · Microsoft Learn Student Ambassador

Participate in IEEE KEC KTM Student Branch, Electronics Project Club, KEC Maker's Space, and KEC Music Club.

Last Updated: August 24, 2026